

Part—II

GTK+ Programming with Glade and Perl

Earlier in this series of articles, we created a small application using Glade and Perl. In this part of the series, we will explore Glade a little further, and use Perl to control how the execution of the application flows.

To begin with, let's return to our GUI design. Use Glade 3 to open the file we created in the previous article in this series. You'll remember that we did not fill in one of the notebook tabs beyond putting in a simple label. We are going to make this tab a 'Search and edit' screen, i.e., the user should be able to search for employee details based on the first and/or the last name and/or the employee ID. Once you get the employee details from the database, you can give an option to edit the details, if you have the privilege to do so.

First select and delete the place-holder label (in the second notebook tab) that we had added last time. Next, create a 5x4 table in the empty space in the notebook tab. In the first column, place two radio buttons (Row 2 and Row 3). This should indicate the employee's gender. To place a *RadioButton* widget, click on the radio button control, and click on the cell where you want it placed (here, R2C1). Go to the *Properties* pane and change the name to *male_rbt* and the label to *Male*. Create another *RadioButton* widget for *Female*. Use the *Label* widget and create labels like *First Name*, *Last Name*, *EmpID*, *Position* and *Address*. Use the *Text Entry* widget as a user input field for all these entries, except for *Position*. For *Position*, you can dynamically create a *ComboBox* and attach it to the table, so do not add any widget in the cell below the *Position* label. Add two *Button* widgets

for *Search* and *Cancel*. Finally, you can click on the menu bar and delete all entries except *File->Quit*.

Keyboard short-cuts and grouping

You might want to add keyboard short-cuts for some of the controls—for example, the *Male* button. To do that, click the *Male* button, scroll down the *Properties* pane till you see the *Use Underline* property. Select *Yes*, and change the label to *_Male*. This lets the user type *Alt-M* to select this radio button.

Since a *Text Entry* field has no visible label, you can attach a *Label* widget's keyboard short-cut to target a *Text Entry*. For example, to use the keyboard short-cut *Alt-R* to place the cursor in the *First Name* text entry field, click the *Label* widget *First Name* (*fn_lbf*) and set the *Use Underline* feature. Change its label text to '*Fi_rst Name*' and click on '*Mnemonic Widget*'. Choose *fn_ent* as the target widget from the list of controls.

Our two radio buttons are currently independently selectable (both *Male* and *Female* can be selected at the same time). This beats the very purpose of having radio buttons (whose selection is, by definition, mutually exclusive). To make them mutually exclusive, they need to be grouped together. To do that, select the *Female* radio button's *Properties*, and click on the *Group* button. Select the *male_rbt* as the group. Figure 1 is an example of the finished screen.